

PERSONAL ENGINEERING COMPENDIUM

My AI Toolkit

The tools I actually use as an AI Engineer

A pragmatic, fully uncropped reference guide detailing the exact software stack, data frameworks, pipelines, and operational tools driving modern AI solutions.

Rohit Bhandari

AI ENGINEER

Stage 1: Foundation

Core programming capabilities, environment configuration, and version control foundations.

Python

Data Structures, OOP, Scripts, Automation

SQL

Queries, Joins, Optimization, Window Functions

Git

Version Control, GitHub, Branching, PRs

VS Code

Extensions, Debugging, Terminal, Jupyter

Stage 2: Data Architecture

Structured manipulation frameworks and cloud storage targets for scalable ingestion.

Pandas

DataFrames, Cleaning, Transformation, Analysis

PostgreSQL

Database, Joins, Indexes, Queries

AWS S3

Cloud Storage, Buckets, Files, Data Lake

Azure Blob

Cloud Storage, Containers, Integration

Stage 3: Orchestration & Workflows

Coordinating distributed tasks and maintaining data flow across multi-cloud environments.



Stage 4: Data Warehousing

From rapid conversational insights to enterprise-scale columnar analytics engines.

Genie — AI-Powered Natural Language Queries

Current tool for fast, conversational data exploration.

Athena — Serverless S3 Query Service

Pay-per-query pricing with direct SQL interface over data lakes.

Redshift — AWS Columnar Analytics Warehouse

Fully managed large-scale data analytics performance.

Synapse — Azure Limitless Analytics Workspace

Dedicated SQL pools, serverless options, and seamless integration.

Stage 5: Compute Decision Matrix

Evaluating compute paradigms for AI pipelines: Databricks vs. AWS Lambda.

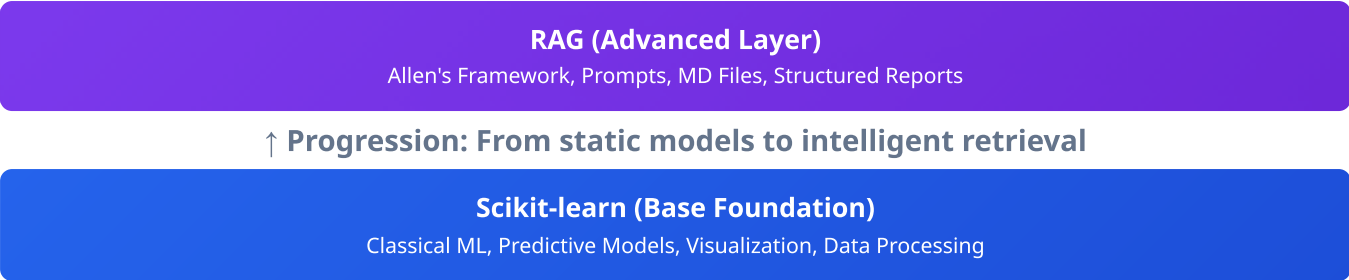
Category	Databricks (Cluster Compute)	Lambda (Serverless Compute)
Use Case	Heavy data processing, distributed machine learning, model training	Quick scripts, event-driven triggers, lightweight tasks
Cost Model	Higher, fixed cluster provisioning overhead	Pay-per-use, highly cost-effective for intermittent tasks
Scalability	Auto-scale multi-node compute clusters	Auto-scale instantly to zero and up concurrently
Complexity	High (requires cluster management and notebook config)	Low (single function deployment and execution runtime)
Best For	Engineering teams building heavy ML pipelines	Serverless event-driven automation and API wrappers

Engineering Takeaway

Choose Databricks when working with massive multi-gigabyte training sets requiring iterative dataframe operations. Opt for Lambda when executing automated monitoring checks or event hooks.

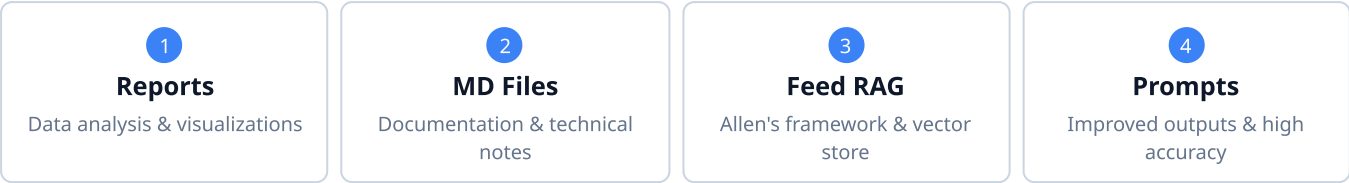
Stage 6: AI & Machine Learning

Bridging traditional classical modeling with intelligent retrieval-augmented generation.



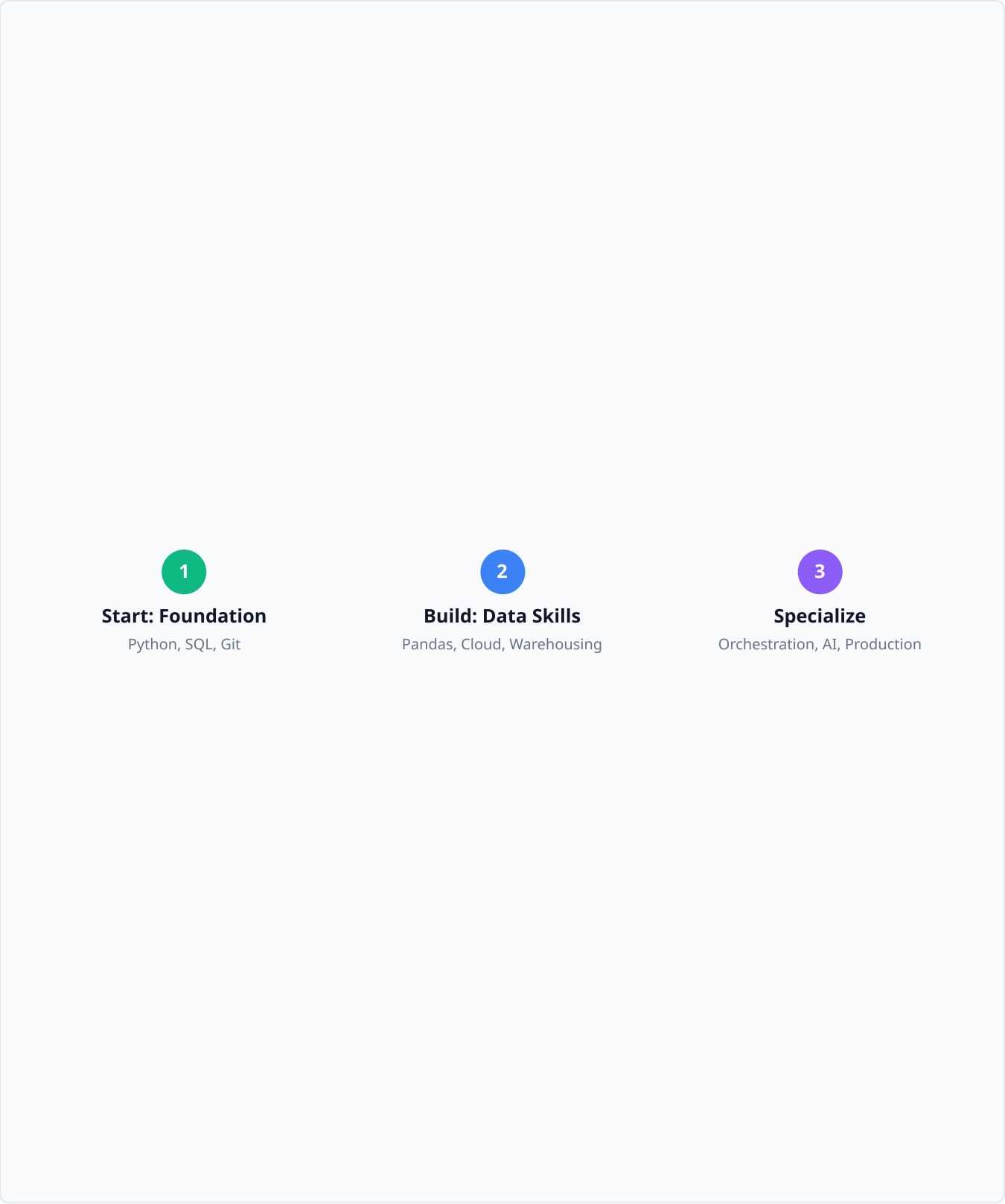
How I Use RAG Workflow

Four-step execution cycle optimizing prompt performance and knowledge retrieval accuracy.



Conclusion: The AI Engineer Journey

An engineer's tool stack evolves across three distinctive evolutionary phases—from raw foundational syntax to enterprise orchestration and AI specialization.



Daily Driver Summary

In practice, my daily routine centers around **Genie** for natural language exploration, **Airflow** for reliable pipeline scheduling, **Scikit-learn** for foundational modeling, and **RAG** workflows configured via Markdown documentation to ensure high-accuracy code generation and prompt responses.